# Native-XOR SAT oracle: pricing its conflicts removes the sub-rho reading

*Derived re-pricing of committed data (ecbench session `ECBS1hb55a1bdabf18`, n = 17 Koblitz, 40 runs per arm). Every number is an unverified assertion [pcc:unverified].*

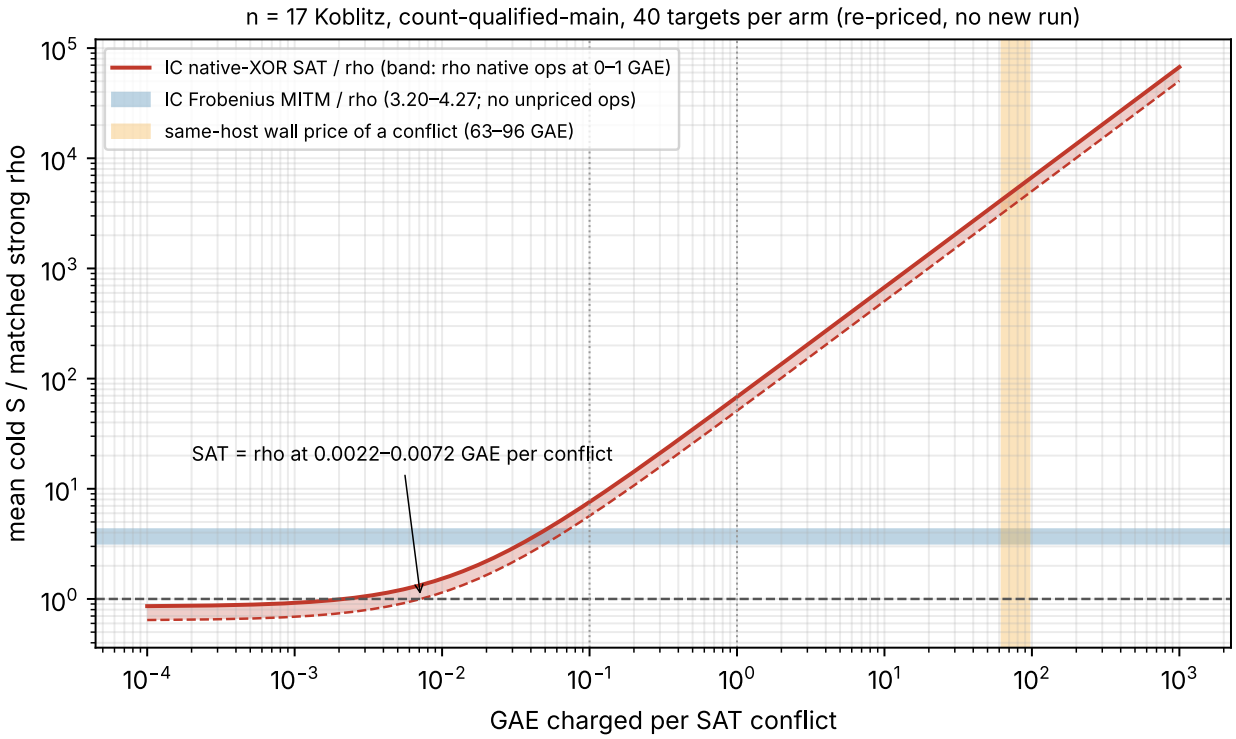



Figure 1: Mean cold S of IC native-XOR SAT over matched strong rho against the GAE charged per CDCL conflict. Rho's unpriced native operations are held at 0 GAE (solid) and 1 GAE (dashed). The orange span is the session's own wall-time price of a conflict.

| w (GAE/op) | SAT S | rho S | SAT / rho | Frob. MITM / rho |
|---|---|---|---|---|
| 0 | 3.41 | 4.00 | 0.854 | 4.27 |
| 0.1 | 30.3 | 4.13 | 7.34 | 4.13 |
| 1 | 272 | 5.34 | 51.0 | 3.20 |

- The order of SAT and rho flips inside the AGENTS.md §5a band, so §5a does not rank the two arms. SAT is below rho only at w = 0.
- **Break-even:** SAT equals rho at 0.0022–0.0072 GAE per conflict. The two values correspond to rho's native operations at 0 and 1 GAE. There are 2,755,966 conflicts, about 68,900 per target.
- **Wall-implied price (descriptive, same host):**
  - 15.1 µs per conflict, against 158–241 ns per GAE, gives 63–96 GAE per conflict;
  - at that price SAT is 3,100–4,800 × rho.
- **Reading:** the sub-rho row was an accounting gap, and the lead closes at toy scale under its own registered criterion. Unit propagations and XOR elimination are not counted, so a per-conflict price is only a floor on solver work.